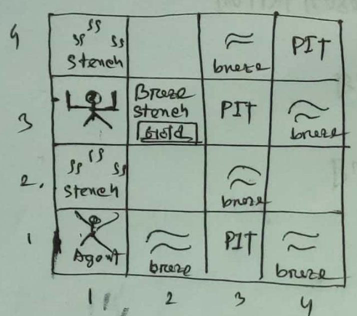
CH – 07



Components:

1. The rooms adjacent to the Wumpus room are smelly.
2. The rooms adjacent to PIT has a breeze.
3. There willbe gliter in the room if and only if the room has gold.
4. The wumpus canbe killed by the agent if the agent is faceing to it.

PEAS description of wumpus world

1. Performance measure
   - +1000 reword point if the agent find the gold.
   - −1000 if agent eatent by wumpus or falling into cave.
   - −1 for each action and −10 for using arrow.

2. Environment:

A 4x4 grid of rooms

- Initially the agent is in the square room [1,1]
- location of gold choose randomly except [1,1]

3. Actuator:

- left turn
- Right turn
- Move forword
- Grab
- Relese
- shoot.

4. sensors:

1. Stench: The agent will perceive the stench if he is in the room adjacent to the Wumpus.
2. Breeze: The agent will perceive breeze if he is in the room adjacent to PIT
3. Gliter: "
4. Bump: The agent ~~the~~ perceive bump if he walks into a wall
5. scream: when the wumpus is shot, emits a horrible scream.

Atomie propositional variable for wumpus world:

let $P_{i,j}$ be true if there is a pit in the room $[i,j]$

$B_{i,j}$

$W_{i,j}$

$S_{i,j}$

$V_{i,j}$

$G_{i,j}$

$OK_{i,j}$

Propositional rule of wumpus world.

(R1) $\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$

(R2) $\neg S_{21} \rightarrow \neg W_{11} \wedge \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$

(R3) $\neg S_{12} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{22} \wedge \neg W_{13}$

(R4) $S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$

Prove that wumpus is in room (1, 3)

① Apply Modus Ponens with $\neg S_{11}$ and R_1

$$\boxed{\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}} \quad \boxed{\neg S_{11}}$$

$$\boxed{\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}}$$

Apply and elemimation rule

$\neg W_{11}$, $\neg W_{12}$ and $\neg W_{21}$

② Apply Modus ponens with $\neg S_{21}$ and R_2

$$\boxed{\neg S_{21} \rightarrow \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}} \quad \boxed{\neg S_{21}}$$

$$\boxed{\neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}}$$

Apply elemimation rule

$\neg W_{21}$, $\neg W_{22}$ and $\neg W_{21}$

Apply Modus ponens of S_{12} and R4

$\boxed{S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}}$ $\boxed{S_{12}}$

$\boxed{W_{13} \vee W_{12} \vee W_{22} \vee W_{11}}$

apply resolution $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$ and $\neg W_{11}$

$\boxed{W_{11} \vee W_{12} \vee W_{13} \vee W_{22}}$ $\boxed{\neg W_{11}}$

$\boxed{W_{12} \vee W_{13} \vee W_{22}}$

Apply resolution $W_{12} \vee W_{13} \vee W_{22}$ and $\neg W_{22}$

$\boxed{W_{12} \vee W_{13} \vee W_{22}}$ $\boxed{\neg W_{22}}$

$\boxed{W_{12} \vee W_{13}}$

Apply resolution $W_{12} \vee W_{13}$ and $\neg W_{12}$

$\boxed{W_{12} \vee W_{13}}$ $\boxed{\neg W_{12}}$

$\boxed{W_{13}}$

Properties of propositional logic

1. Satisfiable : A atomic propositional formula is satisfiable if there is an interpretation for which it is true.

2. Tatulogy : If declarative statement always true

3. Contradiction : If declarative statement is always false.

4. Contingent : A propositional logic can be contingent means if can be neither a tatutology nor a contradiction.

CH-08

FOL Composed of
1. Object → people, number, color
2. Relation → unary → red, round binary → bro. sis
3. Function → Father, best frient.

CH-08

First order logic / Predicate logic

Difference between propositional logic and First order logic.

Propositional logic	First order logic
1. Propositional logic deals with the simple declarative proposition	1. First order logic covers predicates and quantification
2. A propositional logic is collection of declarative statement either true or false	2. Predicate logic is an expression of one or more variable defined on some specific domain
3. Semanties → Truth table	3. Semanties → Interpretation
4. Ex $(P \wedge Q) \rightarrow R$	4. Example $\forall_x [M(x) \longrightarrow P(x)]$
P = It is hot Q = It is humid R = It is rainy "If it is hot and humid, then it is raining"	all man is mortal $M(x)$ = Man $P(x)$ = Mortal

Ans to the Question No- 8.9

1. Emily is either a surgeon or a lawyer

⇒ occupation (Emily, Surgeon) ∨ Occupation (Emily, lawyer)

2. Jon is an actor, but he hold another job.

⇒ occupation (Jon, Actor) ∧ [occupation (Jon, Doctor) ∨ occupation (Jon, Surgeon) ∨ occupation (Jon, Lawyear)

3. All surgeons are doctors

$\Rightarrow \forall_x$ [occupation (x, Surgeon) ⇒ occupation(x, Doctor)

4. Jeo doesnot have a layer (i.e. Jeo is not a customer of any lawyer)

$\Rightarrow \forall_p$ [(occupation (P, lawyer) ⇒ ¬ cowstomer(Jeo, P)]

5. Emily has a boss who is lawyer

⇒ ~~occ~~ $\exists_p$ [Boss (P, Emily) ∧ occupation (P, Layer)]

6.

$\exists_{P_1}$ occupation (P_1, lawyer) ∧ [customer(P_2 P1) ⇒ occupation(P2, doctor)]

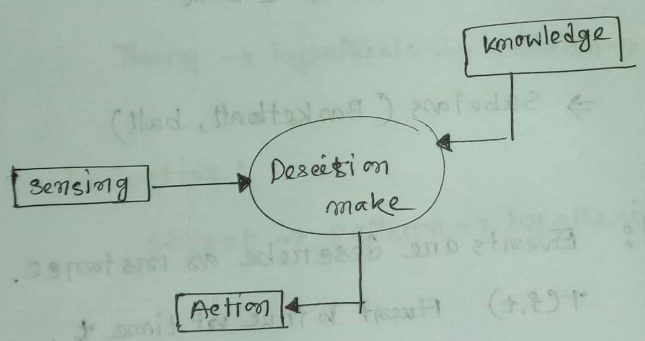
7. Every Surgeon has a lawyer

$\forall_x$ occupation$(x, \text{serjon}) \Rightarrow [\exists_p$ customer$(x, p) \Rightarrow$

occupation$(P_1, \text{Lawyer})]$

CH-12

knowledge representation: is the field of artificial intelligence dedicated to representing information about the world in a form of computer system can use to solve complex task.



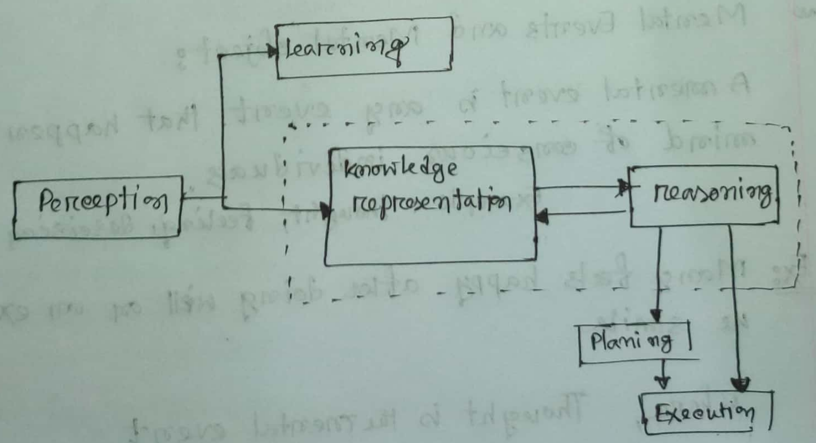


Fig: knowledge cycle

* Catagory and object:

- Catagory also serve to make prediction about objects

- object: An object is a member of catagory

$BB_9 \in Basketball$

- Catagory: Catagory is a subclass of another catagory.

$Basketball \subset Ball$

⇒ Subclass (Basketball, ball)

* Event: Events are describe as instance.

$T(f,t)$ Fluent is true at time t.

* Mental Events and Mental object:

A mental event is any event that happen within mind of conscious individuals.

Example: Thought, feeling, descision

Ex: Many feels happy after doing well on an exam and he smile

where, Thought is the mental event
Smile is the physical event.

Reasoning system for catagories: Reasoning is the mental process of making prediction from available knowledge, facts. It is general process of thingking reationally to find valid conclusion.

Types of reasoning

1. Deductive:

Theory → hypothesis → pattern → confirmation

2. Inductive:

object → pattern → hypthesis → theory.

CH - 22

Natural Language processing: Natural Language processing is the method of communicating with intelligent system.

required:

when we want an intelligent system (robot) to perform as per our instruction.

- I/P or O/P can be
  1. Text
  2. speech

- Components of NLP

① Natural language understanding (NLU)

I/P natural language → useful representation

② Natural language Generation (NLG)

ⓐ Text planning: retrieving relevant content

ⓑ Sentence planning:
- choose required word
- forming useful phrases
- setting tone of the sentence

ⓒ Tex realization: Sentence structure

Difficulties of NLU

1. Logical Ambiguty
2. Syntax level ~~An~~ ambiguty
3. Referential ambiguty.

Stepes in NLP

1. Logical Analysis
2. Syntactic Analysis
3. Symantic Analysis
4. Discourse Integration
5. Pragmatic Analysie

N- gram character model:

1 - gram $\rightarrow$ Unigram

2 - gram $\rightarrow$ Bigram

3 - gram $\rightarrow$ Trigram

ultimately a written text is composed of characters - letters, digits, puntuation etc. The simplest language model is probability distribution over sequence of characters.

It is also called n-gram model

$$P(c_i \mid c_{1:i-1}) = P(c_i \mid c_{i-2:i-1})$$

$$P(c_{1:N}) = \prod_{i=1}^{N} P(c_i \mid c_{1:i-1}) = \prod_{i=1}^{N} P(c_i \mid c_{i-2:i-1})$$

Where language identification

$$= \arg\max_{l} P(l \mid c_{1:N})$$

$$= \arg\max_{l} P(l)\, P(c_{1:N} \mid l)$$

$$= \arg\max_{l} P(l) \prod_{i=1}^{N} P(c_i \mid c_{i-2:i-1}, l)$$

Text classification: Text classification is the Process to catagoriezing text into organized groups.

Information Retrieval: It is the task of finding document that are relevant to a user's need for information.

Information Retrieval canbe characterized by:

1. A corpus of documents
2. Queries posed in query language
3. A result set
4. A porsentation of result set.

Page Rank Algorithm-

Function HITS (query) returns page with hub and authority numb.

Page ← Expand-Page(Releveng page (query))
For each p in pages do
P. Authority ← 1
P. Hub ← 1
repeat until convergence do
For each p in page do .

$P.\text{Authority} \leftarrow \sum_i \text{inlink}(P).\text{Hub}$

$P.\text{Hub} \leftarrow \sum_i \text{outlink}(P).\text{Authority}$

Normalize(page)

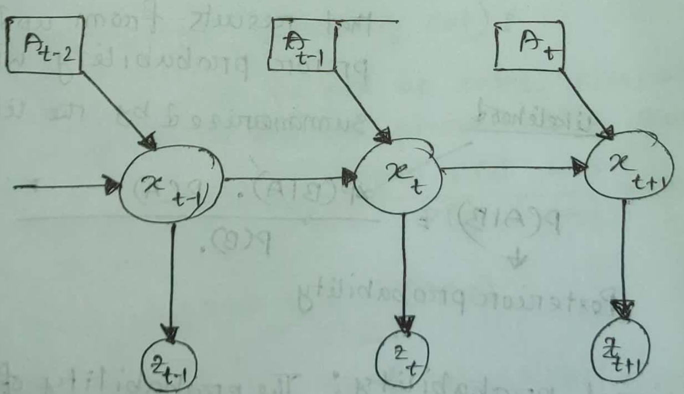
return pages.

CH - 25

Robots: Robots are physical agents that perform task by manipulating the physical world.

Robotic Perception:

Perception is the process by which robot measurement internal representation of the environment.



robotic perception canbe viewed as temporal inference from sequence of action and measurements.

uncertainity

uncertainty : uncertainity is defined as the lack of exat information or knowledge that helps us to find correct conclution.

[unconditional]
* Prior Probability : The probability of an event before new data is collect.

$$P(A) = \frac{\text{No. of desired event}(A)}{\text{Total no of events occures}}$$

[conditional]
* posterior probability : Posterior probability is a Type of conditional probability that results from updating the prior probability with information summarized by the likelihood.

likelihood

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Prior probability

Posterior probability

* Conditional probability : The probability of an event (A), given that another event (B) has already occured.

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}.$$

☐ Probability → uncertainity : came from random variable

* ☐ Random variable → value come from random experime

☐ Marginal probability: The probability of an even occuring (P(A)). It may be thotght of as an unconditional probability.

* ☐ Joint probability: calculate the probability of two events occuring together and at the same point in time.

$$P(A \wedge B) = P(A) * P(B).$$

** ☐ Marginalization: (summing out):
Tells us just to add up some probabilitgies to get the desired probabilistic quantity.

$$\times\ P(a|b) = \frac{P(b|a) . P(a) \rightarrow}{P(b) \rightarrow} \Big\} \text{marginalization}$$

$$P(Y) = \sum_{z \in Z} P(Y, z).$$

☐ Decision Theory: A set of quantitative method for reaching optimal decision.

⊞ Sample space : is a set of possible outcome

⊞ Event : An event is any subset of sample space

⊞ probability distribution: gives a probability for every possible value .

⊠

product rule $P(A \wedge B) = P(A|B)\, P(B)$ . . . . ①

$P(A \wedge B) = P(B|A) \cdot P(A)$ . . . ⑪

Combination of (1)&(2) $P(A|B) = \dfrac{P(B|A)\ P(A)}{P(B)}$ . . . ③

⊞ chain Rule:

$P(A \wedge B) = P(A|B) \cdot P(B)$ ... product rule

$P(A, B) = P(A|B) \cdot P(B)$.

$P(A, B, C, \ldots \ldots n)$

$P(x_1, \ldots, x_n)$

$= P(x_n | x_1, \ldots, x_{n-1})\, P(x_1, \ldots, x_{n-1})$

$= P(x_n | x_1, \ldots, x_{n-1})\, P(x_{n-1} | x_1, \ldots, x_{n-2})\, P(x_1, \ldots$

$= \prod_{i=1}^{n} P(x_i | x_1, \ldots, x_{i-1})$

Construct Naive Bayes Model

we know conditional rule

$$P(a|b) = \frac{P(a \wedge b)}{P(b)}.$$

∴ product rule $P(a \wedge b) = P(a|b) \times P(b)$.

$P(a \wedge b) = P(b|a) \times P(a)$.

Bayes rule $P(a|b) = \frac{P(b|a) \cdot P(a)}{P(b)}$.

The probability distribution form

$$P(a|b) = \frac{P(b|a)\, P(a)}{P(b)}$$

$$= \alpha P(b|a) \cdot P(a). \quad \left[\frac{1}{P(b)} = \alpha\right]$$

Conditional independence

$P(A, B, C)$

$= P(A \mid B, C)\, P(B, C)$

$= P(A \mid B, C)\, P(B|C)\, P(C)$

$= P(A|C)\, P(B|C)\, P(C)$.

use Bayes rule and conditional independence

$P(A|B \wedge C)$

$= P(B \wedge C|A)\, P(A) / P(B \wedge C)$

$= \alpha\, P(B \wedge C|A)\, .\, P(A).$

$= \alpha\, P(B|A)\, P(C|A)\, P(A).$

where. A = ~~effect~~ cause

B, C = effect

$P(\text{cause}, \text{effect}, \text{effect}_n) = P(\text{cause}) \prod_i P(\text{eff} | \text{cause})$

$V_{NB} = \arg\max P(V_j) \prod_i P(a_i | V_j)$

A naive Bayes Model: is a malthematical model that assume the effect are conditionally independent.

classification: Classification is a from of data analysis that extracts model describing data classes.

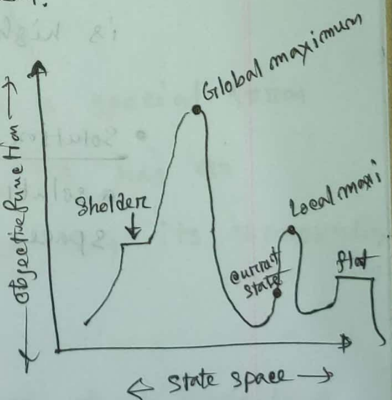
CH-04

Local Search: local search widely used for very big problems. That returns good but not optimal in general.

local search ⟶ 1. Hill climbing.
2.
3.

Hill climbing: It is a local search algorithm, which continuously moves in the direction of increasing value to find the peak of the mountain or best solution to the problem.

Features of Hill climbing
1. Generate and test varient
2. Greedy approach
3. No back tracking



✓ local maximum: local maximum is a state which is better than its neighbor states.

✓ Global maximum: The best possible state of states space.

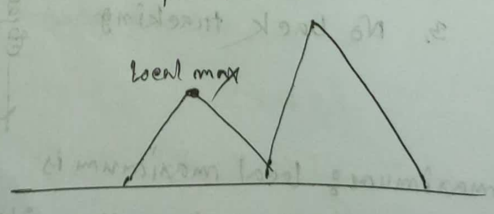
✓ Current state: an agent is currently present flat local maximum :

sholder : It is the plateau region which has an up hill edge.

Problems in Hill climbing.

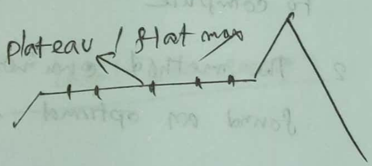
1. local Maximum : A local maximum is a peak state in the landscape which is better than each of its neighboring state, but there is another state also present which is higher than the local maximum.

- Solution : Back tracking technique can be a solution of the local maximum in state space landscape.



2. plateau: A plateau is the flat area of the search space in which all the neighbore states of the current state contain the same value.

- solution: The solution for the plateau is to take big steps or very little steps while searching, to solve the problem.



3. Ridges: A ridge is ~~the~~ a special form of the local maximum. It has an area which is higher than its surrounding areas,

Solution: with the use of bidirectional search, or by moving in different directions we can improve this pr

A Simulated Anneling Search

Advantages:

1. Gives a good solution
3. statically gurrantees finding an optimal solution.

Disadvantage

1. very slow, cost function is expensive to compule
2. The method cannot tell where it has found an optimal solution or not.

A simulated anneling algorithm is a method to solve bound-constrained and unconstrained optimization problem ~~math~~ models

A global optimum solution is generated with simulated annealing.

Hill	Simulated
1. find local optimal	① Find Global optimal
2. No back tracking	② back tracking
3. No gurantee to find best solution	③ find optimal solution
4. stuck 22%. on	④ stuck 2%. m

Local beem search

Genetic Algorithms:

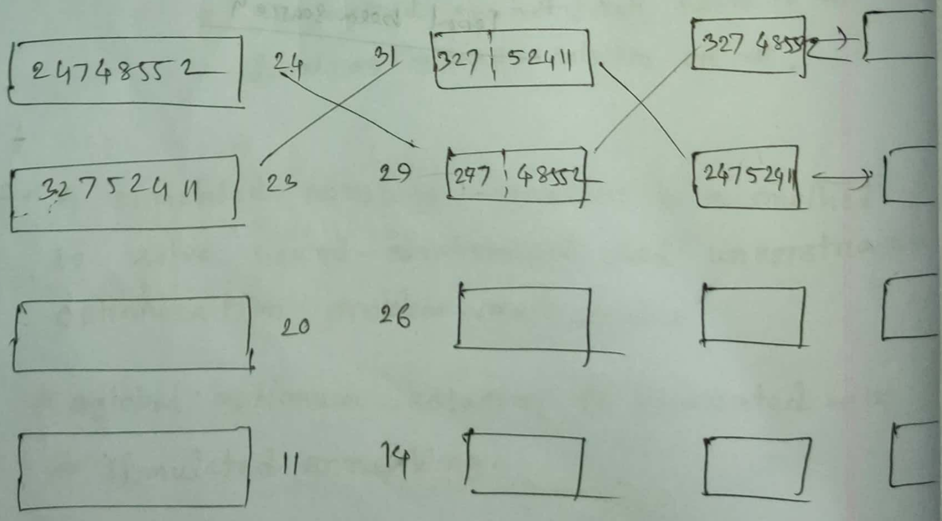
1. Selection
2. crossover
3. mutation

1. Initialize
2. select according to fitness
3. crossover
4. mutation
5. repeat cycle untill true

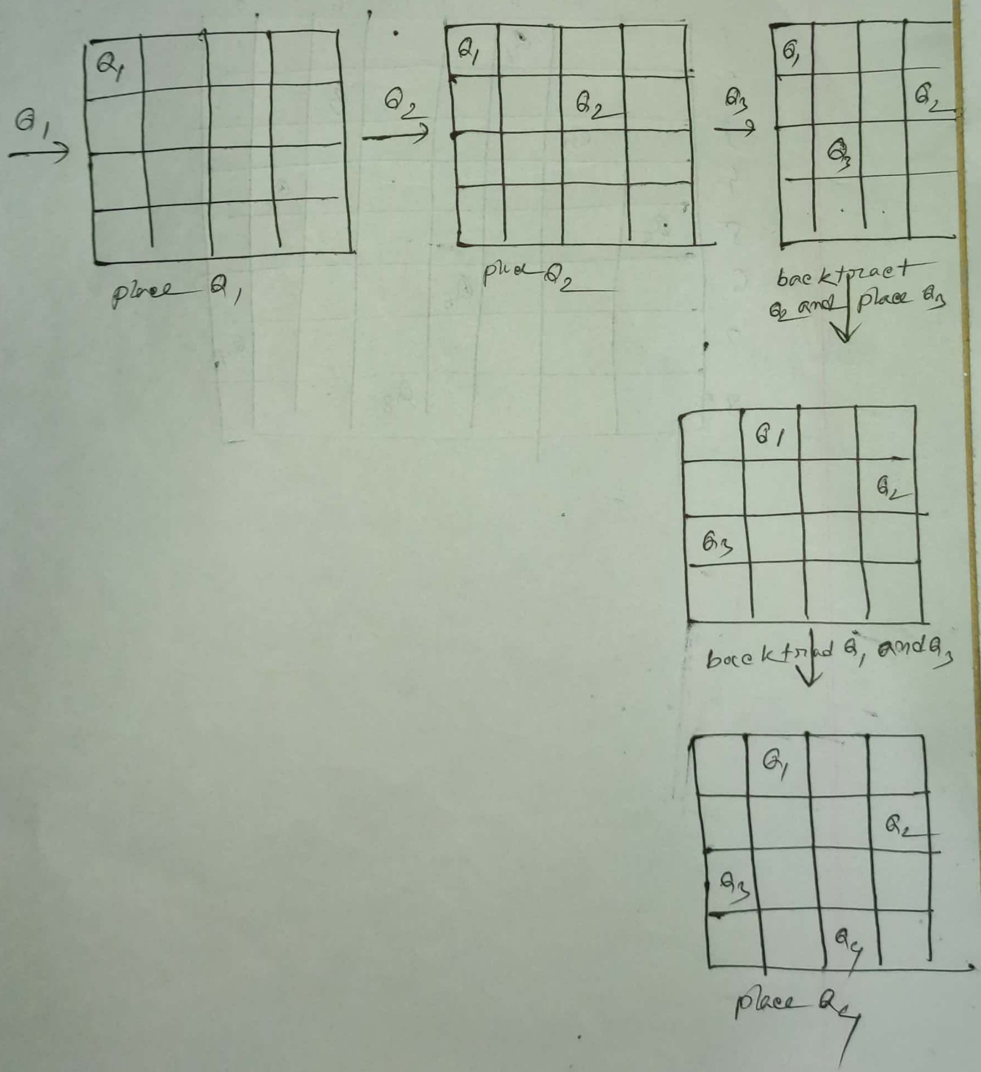
false

stop

step 1	step 2	step 3	step4
24748552	32752411	24415124	32543213
non attacin pairs 24	23	20	4

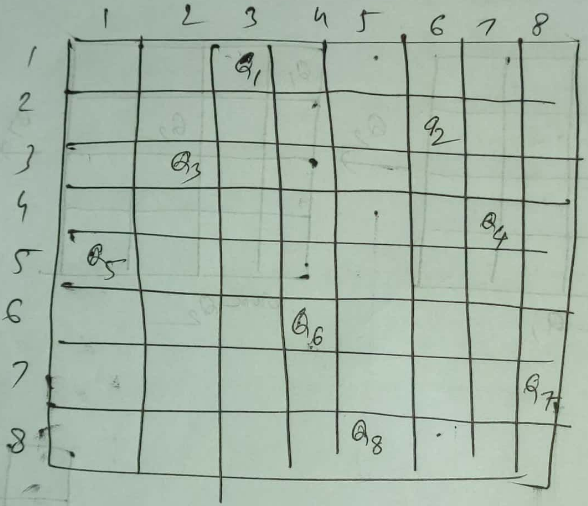


N-Queen problem



For 8 Queen

3,6
2,4



q_1 q_2
36 27 14 85